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European Journal of Organic Chemistry

Communications
doi.org/10.1002/ejoc.202100304

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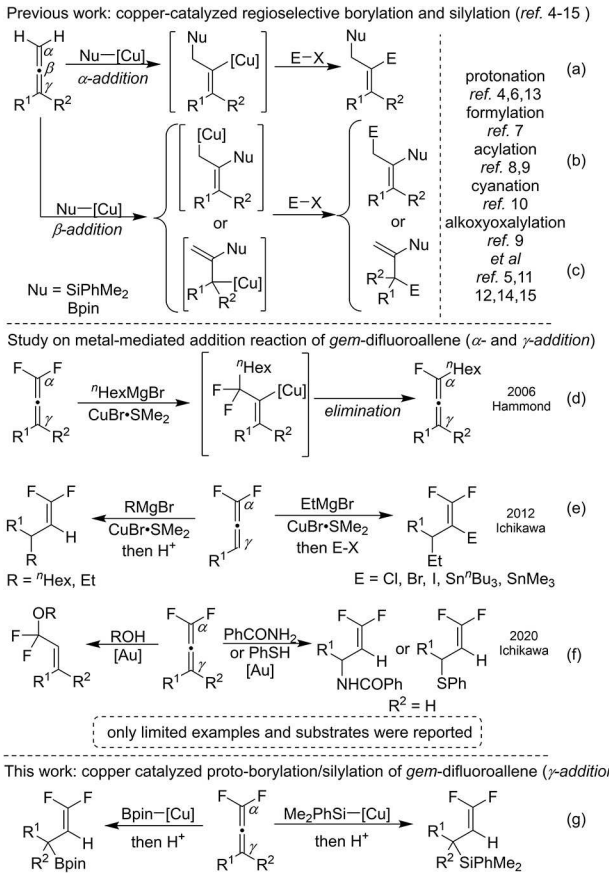
Copper Catalyzed Protosilylation/Protoborylation of *gem*-Difluoroallenes

Cui-Cui Shan,[a] Kai-Yang Dai,[a] Meng Zhao,[a] and Yun-He Xu*[a]

We have developed a protocol for protoborylation and protosilylation of *gem*-difluoroallenes to afford *gem*-difluoroallyl-boronates and *gem*-difluoroallylsilanes. It is different from non-fluorine allenes in which nucleophiles are added to γ-position of *gem*-difluoroallenes to afford β,γ-addition products. Not only monosubstituted *gem*-difluoroallenes but also disubstituted *gem*-difluoroallenes could react well with silaboronate ($PhMe_2Si$–Bpin) and bis(pinacolato)diboron (B_2pin_2). The protocol is simple and efficient to construct compounds which contain tertiary/secondary *gem*-difluoroallylboronates and *gem*-difluoroallylsilanes. Moreover, diverse organofluorine compounds can be converted from these organoboronates and silanes.

In the past few decades, allenes have been shown to demonstrate nice reactivity and selectivity.[1] It can usually be tuned by the electronic or steric effects of its special structure and the nature of the catalysts in reaction conditions. In view of the practicability of these approaches, addition reactions between allene and $PhMe_2Si$–Bpin or B_2pin_2 are investigated in some reviews.[2,3] The different reaction sites of nucleophilic addition between non-fluorine allene and *gem*-difluoroallene by copper catalyst are shown in Scheme 1. Organocopper compounds are added across the terminal double bond of the non-fluorine allene to generate a vinylcopper species, which then reacts with various electrophiles to afford the desired products[4,5] (Scheme 1a). Besides, the nucleophiles added to the β-position of non-fluorine allene generates allylcopper species (Scheme 1b and Scheme 1c). The intermediates react with electrophiles to afford two kinds of β-addition products.[4,5b,6–15]

Substituted *gem*-difluoroallenes are designed to prepare fluorine-containing organic compounds. It is well known that the fluorine-containing organoboron and organosilicon compounds are attractive synthons that have great potentials for the construction of diverse organofluorine compounds.[16] Unlike non-fluorine allene, the organocopper species normally attack the α-position of substituted *gem*-difluoroallenes and yield the monofluoroallene conpounds. Based on the reaction



Scheme 1. α, β and γ-addition of allenes.

pathway, Hammond and collaborators have reported an article involving the reaction of *gem*-difluoroallene with Grignard reagents[17] (Scheme 1d). The difluoroallene underwent an addition-elimination process to yield a trisubstituted monofluoroallene. Besides α-addition, nucleophilic addition of an organocopper species at the γ-position of substituted 1,1-difluoroallenes gave 2,2-difluorovinylmetal intermediates. Capturing of these intermediates by electrophiles (E-X) would provide the desired difluoroalkenes. The *gem*-difluoroalkenes bore newly introduced substituent on the γ-C (Nu) position (Scheme 1e). In 2012, Ichikawa group reported a method for γ-selective addition to 1,1-difluoroallenes using organocopper reagents.[18] A catalytic amount of $CuBr{\cdot}SMe_2$ was not effective in their reaction and the substrates were not well tolerated because Grignard reagents were added. In 2020, the same group reported Au catalyzed α- and γ-addition to *gem*-

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Supporting information for this article is available on the WWW under https://doi.org/10.1002/ejoc.202100304

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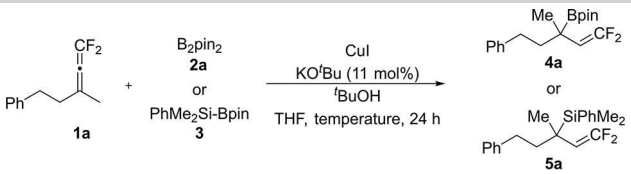
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difluoroallenes which constructed new C–N and C–S bonds on the γ-position[19] (Scheme 1f).

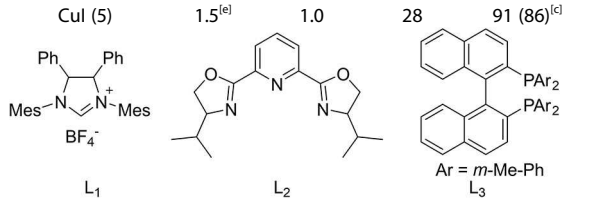
Among the investigation of the reaction between allenes and B_2pin_2 or $PhMe_2Si$–Bpin, we can easily find that neither γ-silylation nor γ-borylation products could be obtained catalyzed by copper catalysts. *Gem*-difluoroallene bears its unique chemical properties, making the reaction positions different from non-fluorine allene. Our group chose substituted *gem*-difluoroallenes as the substrate to react with B_2pin_2 and $PhMe_2Si$–Bpin (Scheme 1g). *Gem*-difluoroallylboronates and *gem*-difluoroallylsilanes could be obtained in our reaction conditions. Nu–[Cu] (silylcopper or borylcopper) species reacting with *gem*-difluoroallene brought about [Cu] connecting the β-carbon atom and Nu connecting the γ-carbon atom. Thus, γ-silylation and γ-borylation of allene are achievable.

In initial attempts, we investigated the reaction conditions of 1,1-difluoroallene **1a** with B_2pin_2 as the model reaction. The reaction was firstly carried out at different temperatures (Table 1, entries 1–4), which generated 86% yield at 50 °C. Next, when different equivalent of ^{t}BuOH was added into the reaction as the source of proton (Table 1, entries 5–6), 87% target product **4a** was obtained with two equivalents of ^{t}BuOH. Furthermore, reducing the amount of B_2pin_2 provided a slight influence on the yield (Table 1, entries 7–8). Similarly, decreased CuI from 10 mol% to 5 mol% had no influence on the reaction (Table 1, entry 9). We tried to add ligand in the reaction to observe how the ligands affected the reaction. When we added 10 mol% PPh_3 to the reaction system, the yield was 87%, which was similar to the reaction result without ligand (Table 1, entry 10). Thus, extra ligands were not needed to add to the reaction. Control experiments showed that no product was observed in the absence of copper catalyst or base (Table 1, entries 11–12). Thus, we can conclude the optimal catalytic conditions to be the following: B_2pin_2 (1.2 equiv), CuI (5 mol%), ^{t}BuOH (2.0 equiv), KO^{t}Bu (11 mol%) and THF (1.0 mL) at 50 °C in argon atmosphere. We then changed B_2pin_2 into $PhMe_2Si$–Bpin to react with 1,1-difluoroallene **1a**. Initially, when the mixture of substrate **1a**, $PhMe_2Si$–Bpin (2.0 equiv), CuI (10 mol%), ^{t}BuOK (11 mol%) and ^{t}BuOH (1.0 equiv) in THF (1.0 mL) reacted at 28 °C, the product **5a** was obtained in 89% yield (Table 1, entry 13). We also tried to add different ligands to the reaction of protosilylation including L_1, L_2, and L_3, and the yields of **5a** were 54%, 14%, and 0%, respectively (Table 1, entries 14–16). It can be seen that the addition of ligand is not conducive to the protosilylation. It had little influence on our reaction when altering the quantity of CuI from 10 mol% to 5 mol% and decreasing the quantity of $PhMe_2Si$–Bpin from 2.0 equiv to 1.5 equiv (Table 1, entries 17–18).

Table 1. Optimization of reaction conditions.[a]



Entry	Cu catalyst [mol%]	R–Bpin [equiv]	^{t}BuOH [equiv]	T [°C]	Yield [%][b]
1	CuI (10)	2.0	1.0	28	72 (65)[c]
2	CuI (10)	2.0	1.0	40	84
3	CuI (10)	2.0	1.0	50	86
4	CuI (10)	2.0	1.0	60	15
5	CuI (10)	2.0	2.0	50	87
6	CuI (10)	2.0	3.0	50	85
7	CuI (10)	1.5	2.0	50	85
8	CuI (10)	1.2	2.0	50	86
9	CuI (5)	1.2	2.0	50	88 (86)[c]
10[f]	CuI (5)	1.2	2.0	50	87
11	–	1.2	2.0	50	N.R.
12[d]	CuI (5)	1.2	2.0	50	N.R.
13	CuI (10)	2.0[e]	1.0	28	93 (89)[c]
14[g]	CuI (10)	2.0[e]	1.0	28	54
15[h]	CuI (10)	2.0[e]	1.0	28	14
16[i]	CuI (10)	2.0[e]	1.0	28	0
17	CuI (10)	1.5[e]	1.0	28	94
18	CuI (5)	1.5[e]	1.0	28	91 (86)[c]



[a] Reaction conditions: **1a** (0.2 mmol), **2**, Cu catalyst, ^{t}BuOH, KO^{t}Bu (11 mol%) and solvent (1.0 mL) in argon, temperature. [b] reactions were determined *via* ^{1}H NMR analysis of crude reaction mixtures. [c] isolated yield. [d] no base. [e] R–Bpin = **3**, $PhMe_2Si$–Bpin, 12 h. [f] PPh_3 (10 mol%). [g] L_1 (12 mol%), KO^{t}Bu (23 mol%). [h] L_2 (12 mol%). [i] L_3 (12 mol%).

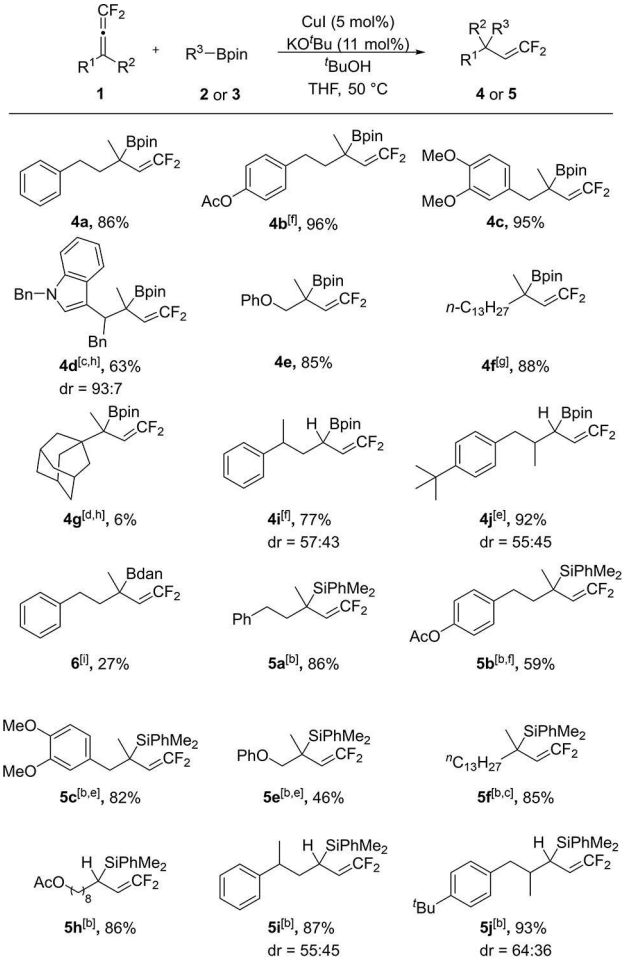
With the optimized conditions in hand (Table 2), we set out to explore the scope of this reaction. As can be seen from Table 2, with ester group on phenyl ring, corresponding product was obtained in 96% yield (Table 2, **4b**). Phenyl ring including two substituent groups afforded the expected products **4c** in 95% yield. The desired product was obtained in 63% yield when phenyl ring was changed to substituted indole (Table 2, **4d**). With *gem*-difluoroallene tether ether or $^nC_{13}H_{27}$ as substrates, we obtained the corresponding products in 85% and 88% yields (Table 2, **4e**, **4f**). From the result of **4g**, we found that sterically hindered substituent greatly reduced the reaction rate. Monosubstituted difluoroallenes reacting with B_2pin_2 gave good to excellent yields (Table 2, **4i**, **4j**). The yield of **4i** was slightly lower than **4j** due to partial loss when purified by silica gel. Substrate **1a** reacting with Bpin–Bdan afforded the corresponding product in 27% yield (Table 2, **6**). The yield was lower than reacting with B_2pin_2.

Besides B_2pin_2 and Bpin–Bdan, *gem*-difluoroallenes also reacted with Me_2PhSi–Bpin at 28 °C (Table 2, **5a**–**5j**). Substrate **1a** reacting with Me_2PhSi–Bpin gave *gem*-difluoroallyl silicon product in 86% isolated yield (Table 2, **5a**). Substrate including ester was well tolerated by the reaction condition (Table 2, **5b**). Substituted benzyl tether difluoroallene reacted well and the desired product was obtained in 82% yield (Table 2, **5c**). Besides, when R^1 was ether or alkyl, desired products **5e** and **5f** were obtained in 46–85% yields. We also observed that monosubstituted difluoroallenes gave higher yields (Table 2, **5h**–**5j**).

Finally, different diboron reagents were also tried for this addition reaction. The corresponding products were further oxidized to *gem*-difluoro-allylalcohols **8** by H_2O_2 because of the instability of products **7** (Table 3, eq 2).

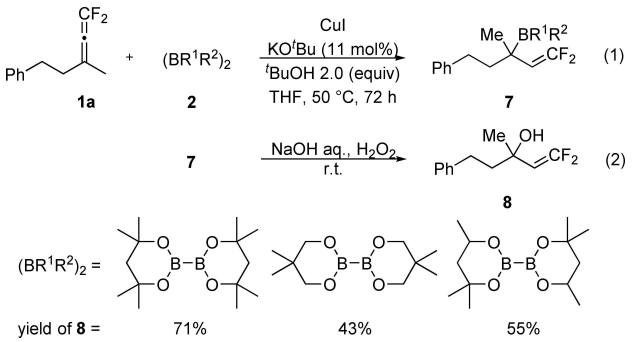
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Table 2. Copper catalyzed addition of *gem*-difluoroallenes.[a]



[a] Reaction conditions: **1** (0.2 mmol), B_2pin_2 (0.24 mmol), CuI (5 mol%), KO ^{t}Bu (11 mol%), ^{t}BuOH (2.0 equiv) and THF (1 mL) in argon, 50 °C for 24 h. The dr values were determined by ^{1}H NMR analysis of crude reaction mixtures. [b] $PhMe_2Si$–Bpin (1.5 equiv), ^{t}BuOH (1.0 equiv), 28 °C, 12 h. [c] CuI (10 mol%). [d] 75 °C. [e] 30 h. [f] 36 h. [g] 48 h. [h] 72 h. [i] Bpin–Bdan (1.2 equiv).

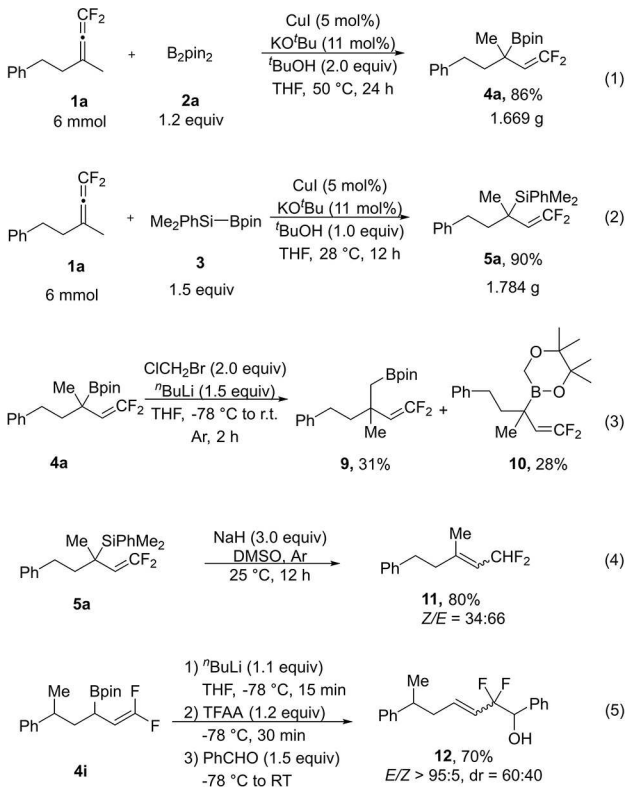
Table 3. Copper catalyzed the addition of *gem*-difluoroallene.[a]



[a] Reaction conditions: **1** (0.2 mmol), **2** (0.24 mmol), CuI (5–10 mol%), KO ^{t}Bu (11 mol%), ^{t}BuOH (2.0 equiv) and THF (1 mL) in argon, 50 °C for 72 h, then, NaOH(aq.), H_2O_2.

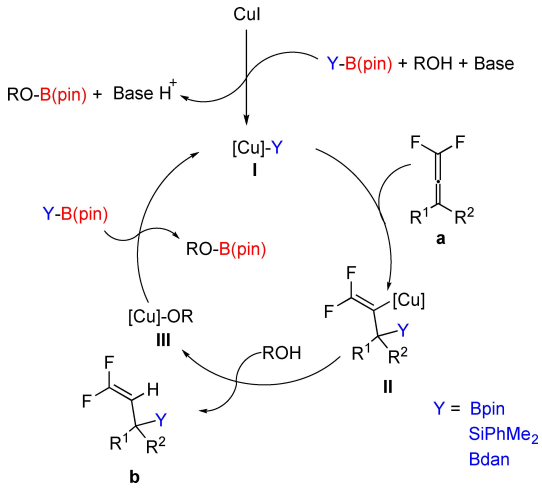
To demonstrate the scalability of our method, we investigated a gram-scale synthesis of *gem*-difluoroallylboronate **4a** and *gem*-difluoroallyl silicon **5a**. When **1a** was carried out on a 6.0-mmol scale, **4a** and **5a** were obtained in 86% and 90% yields (Scheme 2, eqs 1–2). As shown in Scheme 2, further derivatization of **4a** and **5a** has been conducted. When the *gem*-difluoroallylboronate **4a** was subjected to a homologation with a halomethyl lithium reagent (Scheme 2, eq 3), the reaction afforded homoallylboronate **9** and oxygen migration product **10** in 31% and 28% yields. When NaH (3 equiv) was added to the solution of **5a** in DMSO, the product **11** was obtained through allyl protonation process in 80% yield (*Z*/*E* = 34:66) (Scheme 2, eq 4). Then we tried the reaction of *gem*-difluoroallylboronate **4i** with benzaldehyde (Scheme 2, eq 5), resulting in 70% yield (*E*/*Z* > 95:5, dr = 60:40) of product **12**.

Based on these observations, a plausible mechanism for the addition of *gem*-difluoroallene is proposed as illustrated in Scheme 3. The catalytic cycle is initiated by formation of **I** from CuI and Y–Bpin. When Y = Bpin, a boryl copper species [Cu]–Bpin is generated from the reaction between a copper alkoxide and B_2pin_2. Similarly to bis(pinacolato)diboron, borylcopper species Cu–Bdan is generated from Cu–OR and a Bpin–Bdan.[20] The formation of Cu–Bdan can be rationally explained by selective interaction between the Lewis acidic Bpin moiety of Bpin–Bdan[21] and the alkoxide moiety of Cu–OR in the σ-bond metathesis step. If $PhMe_2Si$–Bpin participate in the reaction, a silyl copper species will be afforded. The copper-Y species added to the double bond of allene result in Y always



Scheme 2. Gram-scale synthesis of **4a** and **5a** and their subsequent derivatizations.

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Scheme 3. Proposed reaction mechanism.

being connected to the substituted carbon atom and forms structure **II**. The selective addition of [Cu]-Y to the double bond farther away from fluorine atom is due to the LUMO of the *gem*-difluoroallene is on the remote double bond of fluorine atom.[18,22] Intermediate **II** abstracting a proton from ROH would generate product **b** and **III**. The [Cu]–Y species **I** is regenerate from the [Cu]–OR species through σ-bond metathesis.

In conclusion, we have developed a protocol of copper-catalyzed boryl- and silyl-addition to *gem*-difluoroallenes, which provides access to *gem*-difluoroalkene compounds. The process utilizes a commercially available copper catalyst and tolerates a range of allenes. For borylation process, we attempted different kinds of diboronate reagents. For silylation process, $PhMe_2Si$–Bpin was used. Corresponding tertiary or secondary *gem*-difluoroallylboronates and *gem*-difluoroallylsilanes were obtained regioselectivities.

Acknowledgements

The authors gratefully acknowledge research support of this work by the funding of the National Natural Science Foundation of China (21871240), the State Key Laboratory of Elemento-organic Chemistry Nankai University (202001), and the Fundamental Research Funds for the Central Universities (WK2060000017).

Conflict of Interest

The authors declare no conflict of interest.

Keywords: *gem*-Difluoroallene · Nucleophilic addition · Protoborylation · Protosilylation · Regioselectivity

[1] For selected reviews on the chemistry of allene, see: a) K. K. Wang, *Chem. Rev.* **1996**, *96*, 207–222; b) R. Zimmer, C. U. Dinesh, E. Nandanan, F. A. Khan, *Chem. Rev.* **2000**, *100*, 3067–3126; c) J. A. Marshall, *Chem. Rev.* **2000**, *100*, 3163–3186; d) A. Hoffmann-Röder, N. Krause, *Angew. Chem. Int. Ed.* **2004**, *43*, 1196–1216; *Angew. Chem.* **2004**, *116*, 1216–1236; e) S. Ma, *Acc. Chem. Res.* **2003**, *36*, 701–712; f) S. Ma, *Acc. Chem. Res.* **2009**, *42*, 1679–1688.

[2] For selected reviews for the reaction between allenes and $PhMe_2Si$–Bpin, see: a) J.-J. Feng, W. Mao, L. Zhang, M. Oestreich, *Chem. Soc. Rev.* **2021**, *50*, 2010–2073; b) M. Oestreich, E. Hartmann, M. Mewald, *Chem. Rev.* **2013**, *1113*, 402–441; c) C. Moberg, *Synthesis* **2020**, *52*, 3129–3139; d) J. R. Wilkinson, C. E. Nuyen, T. S. Carpenter, S. R. Harruff, R. V. Hoveln, *ACS Catal.* **2019**, *9*, 8961–8979.

[3] For selected reviews for the reaction between allenes and B_2pin_2, see: a) W. Xue, M. Oestreich, *ACS Cent. Sci.* **2020**, *6*, 1070–1081; b) K. Semba, T. Fujihara, J. Terao, Y. Tsuji, *Tetrahedron* **2015**, *71*, 2183–2197; c) Y. Tsuji, T. Fujihara, *Chem. Rec.* **2016**, *16*, 2294–2313; d) A. P. Pulis, K. Yeung, D. J. Procter, *Chem. Sci.* **2017**, *8*, 5240–5247; e) C. Nájera, I. P. Beletskaya, M. Yus, *Chem. Soc. Rev.* **2019**, *48*, 4515–4618; f) T. Fujihara, Y. Tsuji, *Synthesis* **2018**, *50*, 1737–1749; g) D. Hemming, R. Fritzemeier, S. A. Westcott, W. L. Santos, P. G. Steel, *Chem. Soc. Rev.* **2018**, *47*, 7477–7494; h) J. Carreras, A. Caballero, P. J. Pérez, *Chem. Asian J.* **2019**, *14*, 329–343; i) F. J. T. Talbot, Q. Dherbassy, S. Manna, C. Shi, S. Zhang, G. P. Howell, G. J. P. Perry, D. J. Procter, *Angew. Chem. Int. Ed.* **2020**. *59*, 20278–20289; *Angew. Chem.* **2020**. *132*, 20454–20465.

[4] For selected examples for borylvinyl copper intermediates generated from allenes and Cu–Bpin species, see: a) K. Semba, M. Shinomiya, T. Fujihara, J. Terao, Y. Tsuji, *Chem. Eur. J.* **2013**, *19*, 7125–7132; b) W. Yuan, L. Song, S. Ma, *Angew. Chem. Int. Ed.* **2016**, *55*, 3140–3143; *Angew. Chem.* **2016**, *128*, 3192–3195.

[5] For selected examples for silylvinyl copper intermediates generated from allenes and Cu–Si species, see: a) H. Yoshida, Y. Hayashi, Y. Ito, K. Takaki, *Chem. Commun.* **2015**, *51*, 9440–9442; b) Y. Tani, T. Fujihara, J. Terao, Y. Tsuji, *J. Am. Chem. Soc.* **2014**, *136*, 17706–17709.

[6] For selected examples for protoborylation through β-borylallyl copper intermediates generated from allenes and Cu–Bpin species, see: a) S. B. Thorpe, X. Guo, W. L. Santos, *Chem. Commun.* **2011**, *47*, 424–426; b) W. Yuan, S. Ma, *Adv. Synth. Catal.* **2012**, *354*, 1867–1872; c) W. Yuan, X. Zhang, Y. Yu, S. Ma, *Chem. Eur. J.* **2013**, *19*, 7193–7202; d) L. Song, W. Yuan, S. Ma, *Org. Chem. Front.* **2017**, *4*, 1261–1265; e) F. Meng, B. Jung, F. Haeffner, A. H. Hoveyda, *Org. Lett.* **2013**, *15*, 1414–1417; f) H. Jang, B. Jung, A. H. Hoveyda, *Org. Lett.* **2014**, *16*, 4658–4661.

[7] For selected an example for boraformylation through β-borylallyl copper intermediates generated from allenes and Cu–Bpin species, see: T. Fujihara, A. Sawada, T. Yamaguchi, Y. Tani, J. Terao, Y. Tsuji, *Angew. Chem. Int. Ed.* **2017**, *56*, 1539–1543; *Angew. Chem.* **2017**, *129*, 1561–1565.

[8] For selected examples for boroacylation through β-borylallyl copper intermediates generated from allenes and Cu–Bpin species, see: a) A. Boreux, K. Indukuri, F. Gagosz, O. Riant, *ACS Catal.* **2017**, *7*, 8200–8204; b) J. Han, W. Zhou, P.-C. Zhang, H. Wang, R. Zhang, H.-H. Wu, J. Zhang, *ACS Catal.* **2019**, *9*, 6890–6895; c) B.-X. Shen, X.-T. Min, Y.-C. Hu, L.-L. Qian, S.-N. Yang, B. Wan, Q.-A. Chen, *Org. Biomol. Chem.* **2020**, *18*, 9253–9260.

[9] For selected an example for bora-acylation and alkoxyoxalylation through β-borylallyl copper intermediates generated from allenes and Cu–Bpin species, see: A. Sawada, T. Fujihara, Y. Tsuji, *Adv. Synth. Catal.* **2018**, *360*, 2621–2625.

[10] For selected an example for cyanoborylation through β-borylallyl copper intermediates generated from allenes and Cu–Bpin species, see: Z. Li, L. Zhang, M. Nishiura, G. Luo, Y. Luo, Z. Hou, *ACS Catal.* **2020**, *10*, 11685–11692.

[11] For selected examples for borylation through β-borylallyl copper intermediates generated from allenes and Cu–Bpin species, see: a) Y. Zhou, W. You, K. B. Smith, M. K. Brown, *Angew. Chem. Int. Ed.* **2014**, *53*, 3475–3479; *Angew. Chem.* **2014**, *126*, 3543–3547; b) F. Meng, K. P. McGrath, A. H. Hoveyda, *Nature* **2014**, *513*, 367–374; c) J. Rae, K. Yeung, J. J. W. McDouall, D. J. Procter, *Angew. Chem. Int. Ed.* **2016**, *55*, 1102–1107; *Angew. Chem.* **2016**, *128*, 1114–1119; d) K. Yeung, R. E. Ruscoe, J. Rae, A. P. Pulis, D. J. Procter, *Angew. Chem. Int. Ed.* **2016**, *55*, 11912–11916; *Angew. Chem.* **2016**, *128*, 12091–12095; e) H. Jang, F. Romiti, S. Torker, A. H. Hoveyda, *Nat. Chem.* **2017**, *9*, 1269–1275; f) K. Yeung, F. J. T. Talbot, G. P. Howell, A. P. Pulis, D. J. Procter, *ACS Catal.* **2019**, *9*, 1655–1661; g) H. Deng, Z. Meng, S. Wang, Z. Zhang, Y. Zhang, Y. Shangguan, F. Yang, D. Yuan, H. Guo, C. Zhang, *Adv. Synth. Catal.* **2019**, *361*, 3582–3587; h) K. Semba, N. Bessho, T. Fujihara, J. Terao, Y. Tsuji, *Angew. Chem. Int. Ed.* **2014**, *53*, 9007–9011; *Angew. Chem.* **2014**, *126*,

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9153–9157; i) T. Song, L. Zhu, H. Li, C.-H. Tung, Y. Lan, Z. Xu, *Org. Lett.* **2020**, *22*, 2419–2424; j) C.-Y. Zhao, H. Zheng, D.-W. Ji, X.-T. Min, Y.-C. Hu, Q.-A. Chen, *Cell Rep. Phys. Sci.* **2020**, *1*, 100067; k) Y. Huang, S. Torker, X. Li, J. del Pozo, A. H. Hoveyda, *Angew. Chem. Int. Ed.* **2019**, *58*, 2685–2691; *Angew. Chem.* **2019**, *131*, 2711–2717; l) Y. Ozawa, H. Iwamoto, H. Ito, *Chem. Commun.* **2018**, *54*, 4991–4994; m) S.-C. Zhang, J. del Pozo, F. Romiti, Y.-C. Mu, S. Torker, A. H. Hoveyda, *Science* **2019**, *364*, 45–51.

[12] For selected examples for the reaction mention allenes and B_2pin_2, see: a) F. Meng, H. Jang, B. Jung, A. H. Hoveyda, *Angew. Chem. Int. Ed.* **2013**, *52*, 5046–5051; *Angew. Chem.* **2013**, *125*, 5150–5155; b) K. Semba, T. Fujihara, J. Terao, Y. Tsuji, *Angew. Chem. Int. Ed.* **2013**, *52*. 12400–12403; *Angew. Chem.* **2013**, *125*, 12626–12629; c) Y. Takemoto, H. Yoshida, K. Takaki, *Synthesis* **2014**, *46*, 3024–3032; d) W. Zhao, J. Montgomery, *J. Am. Chem. Soc.* **2016**, *138*, 9763–9766; e) Y.-S. Zhao, X.-Q. Tang, J.-C. Tao, P. Tian, G.-Q. Lin, *Org. Biomol. Chem.* **2016**, *14*, 4400–4404; f) F. Meng, X. Li, S. Torker, Y. Shi, A. H. Hoveyda, *Nature* **2016**, *537*, 387–393; g) S. Akiyama, S. Nomura, K. Kubota, H. Ito, *J. Org. Chem.* **2020**, *85*, 4172–4181; h) K.-R. Feng, Y.-X. Tan, W. Ye, Y.-F. Wang, Z.-T. He, P. Tian, G.-Q. Lin, *Org. Lett.* **2021**, *23*, 607–611.

[13] For selected examples for proto-silylation through β-silylallyl copper intermediates generated from allenes and Cu–Si species, see: a) Y.-H. Xu, L.-H. Wu, J. Wang, T.-P. Loh, *Chem. Commun.* **2014**, *50*, 7195–7197; b) J. Rae, Y. C. Hu, D. J. Procter, *Chem. Eur. J.* **2014**, *20*, 13143–13145; c) S. Pashikanti, J. A. Calderone, M. K. Nguyen, C. D. Sibley, W. L. Santos, *Org. Lett.* **2016**, *18*, 2443–2446.

[14] For selected an example for silylative allylation of ketones and aldehydes through β-silylallyl copper intermediates generated from allenes and Cu–Si species, see: Y. Tani, T. Yamaguchi, T. Fujihara, J. Terao, Y. Tsuji, *Chem. Lett.* **2015**, *44*, 271–273.

[15] For selected an example for silacarboxylation through β-silylallyl copper intermediates generated from allenes and Cu–Si species, see: Z.-T. He, X.-Q. Tang, L.-B. Xie, M. Cheng, P. Tian, G.-Q. Lin, *Angew. Chem. Int. Ed.* **2015**, *54*, 14815–14818; *Angew. Chem.* **2015**, *127*, 15028–15031.

[16] For selected examples for the synthesis and applications of fluoro-containing compounds, see: a) R. Berger, G. Resnati, P. Metrangolo, E. Weber, J. Hulliger, *Chem. Soc. Rev.* **2011**, *40*, 3496–3508; b) T. Fujiwara, D. O'Hagan, *J. Fluorine Chem.* **2014**, *167*, 16–29; c) E. P. Gillis, K. J. Eastman, M. D. Hill, D. J. Donnelly, N. A. Meanwell, *J. Med. Chem.* **2015**, *58*, 8315–8359; d) J. Wang, M. Sánchez-Roselló, J. L. Aceña, C. del Pozo, A. E. Sorochinsky, S. Fustero, V. A. Soloshonok, H. Liu, *Chem. Rev.* **2014**, *114*, 2432–2506; e) J. Ichikawa, T. Sonoda, H. Kobayashi, *Tetrahedron Lett.* **1989**, *30*, 1641–1644; f) O. A. Argintaru, D. Ryu, I. Aron, G. A. Molander, *Angew. Chem. Int. Ed.* **2013**, *52*, 13656–13660; *Angew. Chem.* **2013**, *125*, 13901–13905; g) W.-H. Guo, H.-Y. Zhao, Z.-J. Luo, S. Zhang, X. Zhang, *ACS Catal.* **2019**, *9*, 38–43.

[17] M. Mae, J. A. Hong, B. Xu, G. B. Hammond, *Org. Lett.* **2006**, *8*, 479–482.

[18] K. Fuchibe, M. Ueda, M. Yokota, J. Ichikawa, *Chem. Lett.* **2012**, *41*, 1619–1621.

[19] K. Fuchibe, M. Abe, M. Sasaki, J. Ichikawa, *J. Fluorine Chem.* **2020**, *232*, 109452.

[20] a) H. Yoshida, Y. Takemoto, K. Takaki, *Chem. Commun.* **2014**, *50*, 8299–8302; b) H. Yoshida, Y. Takemoto, K. Takaki, *Asian J. Org. Chem.* **2014**, *3*, 1204–1209; c) H. Yoshida, Y. Takemoto, K. Takaki, *Chem. Commun.* **2015**, *51*, 6297–6300.

[21] a) H. Noguchi, K. Hojo, M. Suginome, *J. Am. Chem. Soc.* **2007**, *129*, 758–759; b) H. Noguchi, T. Shioda, C.-M. Chou, M. Suginome, *Org. Lett.* **2008**, *10*, 377–380; c) N. Iwadate, M. Suginome, *Chem. Lett.* **2010**, *39*, 558–560.

[22] L. N. Domelsmith, K. N. Houk, C. Piedrahita, W. J. Dolbier, Jr., *J. Am. Chem. Soc.* **1978**, *100*, 6908–6911.

Manuscript received: March 13, 2021
Revised manuscript received: April 10, 2021
Accepted manuscript online: April 19, 2021

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